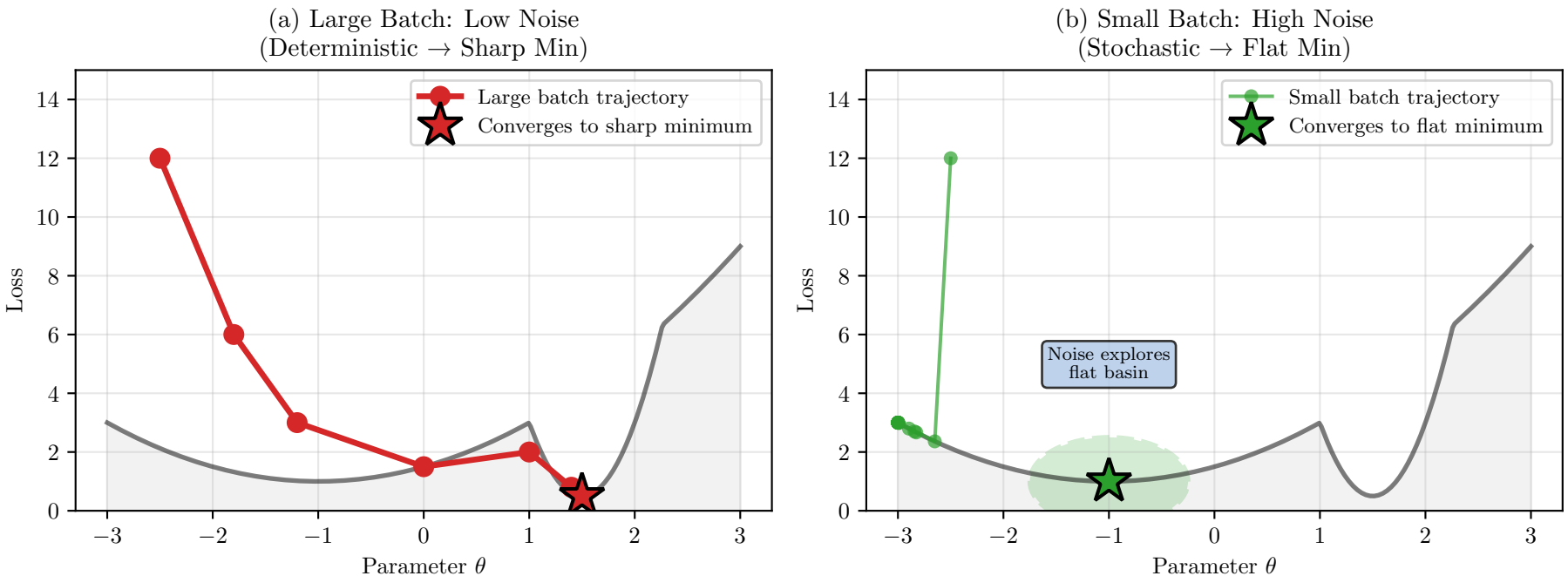
SGD Noise as Implicit Regularization Toward Flat Minima
(a) Large Batch: Low Noise
(Deterministic → Sharp Min)
Large batch trajectory
Converges to sharp minimum
Loss
Parameter θ
(b) Small Batch: High Noise
(Stochastic → Flat Min)
Small batch trajectory
Converges to flat minimum
Noise explores
flat basin
Loss
Parameter θ